Q1.

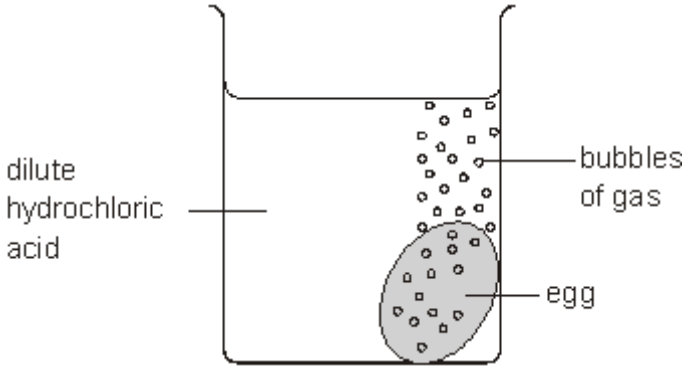
(a) The table below shows the pH of four acidic liquids.

acidic liquid	pH
grapefruit juice	3.1
ethanoic acid	3.0
lemonade	4.4
dilute hydrochloric acid	1.0

Which of these liquids is the **least** acidic?

1 mark

(b) Emilio cooked an egg until it was hard-boiled.
He put the egg in a beaker of dilute hydrochloric acid as shown.



(i) The egg shell reacted completely with the acid.
After two days the pH of the liquid in the beaker was 2.5.

How did the **acidity** of the liquid in the beaker change?
Use the table above to help you.

.....

1 mark

(ii) Emilio put another hard-boiled egg in some ethanoic acid.
It took longer for the shell to react completely.

Use the table opposite to suggest a reason for this.

.....

.....

1 mark

(c) The chemical formulae for four acids are shown in the table below.

sulphuric acid	**hydrochloric acid**	**nitric acid**	**ethanoic acid**
H_2SO_4	HCl	HNO_3	CH_3COOH

(i) Give the **name** of the element that is present in all four acids.

.....

1 mark

(ii) Give the **names** of the two **other** elements present in sulphuric acid.

1. .....

1 mark

2. .....

1 mark

(iii) How many atoms are there in the formula HNO_3 (nitric acid)?

.....

1 mark

maximum 7 marks

Q1.

(a) lemonade

accept '4.4'

1 (L5)

(b) (i) any **one** from

- it is less acidic

  accept 'it is weaker'
  *do **not** accept 'it has become more alkaline'*
  accept 'it decreases'

- its pH has increased

  accept 'it has gone from 1.0 to 2.5'
  *do **not** accept 'it increased'*
  *do **not** accept 'the acidity changed by pH 1.5'*

1 (L5)

(ii) any **one** from

- ethanoic acid is less acidic than the hydrochloric acid

  accept the converse
  'ethanoic acid is pH 3' is insufficient
  accept 'it is less acidic'
  'ethanoic acid is weak' is insufficient

- ethanoic acid has a higher pH

  accept 'partially neutralised'
  'it is weaker' is insufficient

- it has a pH of 3 compared with 1

- it is a weaker acid

1 (L6)

(c) (i) hydrogen

*'H_2' **or** 'H' are insufficient*

1 (L6)

(ii) • sulphur

1 (L6)

- oxygen

1 (L6)

*'S' **or** 'O' **or** 'O_2' are insufficient*
answers can be in either order

(iii) 5

1 (L6)

[7]